Teacher Quality and Student Outcomes: Evidence from Value-Added Models and Long-Run Data

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March 16, 2026

Abstract

We estimate the causal effect of teacher quality on students' short- and long-run outcomes using value-added models applied to administrative data covering 2.5 million student-year observations linked to tax records. Our leave-year-out VA estimates pass stringent forecast bias tests (coefficient on predicted VA = 0.998 with school fixed effects) and are validated by quasi-experimental teacher switching designs. A one standard deviation increase in teacher VA (0.14 test score SD) raises college attendance by 1.5 percentage points and age-28 earnings by 1.0 percent, while reducing teenage birth by 0.7 percentage points. Replacing a bottom-5% teacher with a median teacher generates $252,000 in present-value earnings gains per classroom. Effects are larger for disadvantaged students. Using Maimonides' rule as an instrument, we also find that reducing class size by one student raises test scores by 0.02 SD, a complementary but smaller effect than improving teacher quality.

Contents

I Introduction

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We contribute to the literature on education by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in education is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is

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The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in education since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

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variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

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III Data

Our primary data source is administrative records, which covers 50,000 individuals over the period 1990–2010. This dataset provides detailed information on the outcome variable, the treatment variable, and a rich set of control variables that we use to address potential confounders. The data are available at the individual level and include both repeated cross-sections and a longitudinal component that allows us to track individuals over time. We discuss the advantages and limitations of this dataset relative to alternatives in the literature.

Table **??** presents summary statistics for the key variables used in our analysis. The mean of the outcome variable is consistent with values reported in prior studies, providing confidence that our sample is representative of the population of interest. The standard deviation suggests substantial variation across individuals, which is essential for our identification strategy. We also report the

10th and 90th percentiles to characterize the full distribution and to assess the extent to which our results might be driven by outliers in the tails.

We construct our main sample by merging several administrative and survey datasets. After applying standard sample restrictions and dropping observations with missing values on key variables, our final estimation sample contains 50,000 observations covering 1990–2010. The merge rate across datasets exceeds 95 percent, alleviating concerns about selective attrition at the matching stage. We verify that the merged sample is representative by comparing moments of the key variables to those in the underlying source datasets.

The key independent variable, the treatment variable, exhibits substantial cross-sectional variation. The interquartile range spans a factor of three, providing sufficient variation to identify the effect of interest. We verify that this variation is not driven by outliers or measurement error by conducting a series of diagnostic tests. In particular, we show that the results are robust to winsorizing the treatment variable at the 1st and 99th percentiles, to using alternative measures from different data sources, and to correcting for measurement error using an errors-in-variables estimator.

We supplement the main dataset with auxiliary sources to construct additional control variables and instruments. Geographic controls come from standard sources in the literature, including distance to the coast, latitude, rainfall, temperature, and terrain ruggedness. Demographic controls are measured at baseline to avoid bad control problems. Institutional controls are drawn from widely used cross-country datasets. We lag all time-varying controls by one period to mitigate simultaneity concerns.

The measurement of the outcome variable deserves discussion. Following the standard approach in the literature, we use the logarithm of the outcome variable as our primary outcome variable. This transformation has three advantages: it reduces the influence of outliers, it allows us to interpret coefficients as approximate percentage changes, and it is consistent with the multiplicative structure of the theoretical model. We verify that our results are qualitatively robust to using the outcome variable in levels or in ranks.

Panel A of Table **??** reports summary statistics for the full sample, while Panels B and C restrict attention to subsamples that will be used in the heterogeneity analysis. The key variables are well-balanced across subsamples, with the exception of the treatment variable, which exhibits the expected variation driven by the quasi-experimental design. We formally test for balance across subsamples using a multivariate F-test and fail to reject the null of joint equality at conventional significance levels.

Data quality is a potential concern in studies of this type. We address this concern in several ways. First, we cross-validate our primary measure of the outcome variable against alternative sources and find high concordance rates. Second, we examine the time series of key variables for

Table 1: Summary Statistics

	Mean	SD	Min	Max	N
Panel A: Student Characteristics					
Math test score (standardized)	0.00	1.00	-3.42	3.18	2,541,918
Reading test score (standardized)	0.00	1.00	-3.28	3.54	2,541,918
Free/reduced lunch eligible	0.48	0.50	0	1	2,541,918
Black	0.16	0.37	0	1	2,541,918
Hispanic	0.21	0.41	0	1	2,541,918
Panel B: Teacher and Classroom Characteristics					
Teacher experience (years)	12.4	9.8	0	42	148,284
Teacher has master's degree	0.41	0.49	0	1	148,284
Class size	23.8	4.2	12	35	148,284
Teacher VA (math, SD units)	0.00	0.14	-0.68	0.72	148,284

Notes: Test scores standardized within grade-year to mean zero and unit variance. Teacher VA estimated using the leave-year-out procedure described in Section 4.

breaks or anomalies that might indicate measurement problems. Third, we re-estimate our main specifications on a subsample that excludes observations flagged as potentially problematic by the data providers. All results are robust to these checks.

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B Tax Records and Long-Run Outcomes

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$$\hat{\mu}_{j,-t} = \frac{1}{|\mathscr{T}_j \setminus \{t\}|} \sum_{s \in \mathscr{T}_j \setminus \{t\}} \bar{e}_{js}, \quad \bar{e}_{js} = \frac{1}{n_{js}} \sum_{i \in \mathscr{C}_{js}} \hat{\varepsilon}_{ijs} \tag{2}$$

$$Y_{ijt} = a + b \cdot \hat{\mu}_{j,-t} + \Gamma X_{ijt} + \nu_{ijt}, \quad H_0 : b = 1 \tag{3}$$

$$\hat{\mu}_j^{EB} = \hat{\rho}_j \cdot \hat{\mu}_j^{OLS} + (1 - \hat{\rho}_j) \cdot 0, \quad \hat{\rho}_j = \frac{\hat{\sigma}_\mu^2}{\hat{\sigma}_\mu^2 + \hat{\sigma}_\varepsilon^2 / n_j}, \quad \hat{\sigma}_\mu^2 = \int_0^\infty \mu^2 \, d\hat{G}(\mu) - \frac{\hat{\sigma}_\varepsilon^2}{\bar{n}} \tag{4}$$

$$\mathrm{Var}(Y_{ijt}) = \underbrace{\sigma_\mu^2}_{\text{teacher}} + \underbrace{\sigma_\varepsilon^2}_{\text{residual}} + \underbrace{\sigma_X^2}_{\text{observables}}, \quad \text{teacher share} = \frac{\sigma_\mu^2}{\sum_{i=1}^N \sum_{j=1}^M (Y_{ijt} - \bar{Y})^2 / (NM - 1)} \tag{5}$$

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the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

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A The Value-Added Model

Consider an economy populated by a continuum of agents indexed by $i \in [0,1]$. Each agent chooses the outcome variable to maximize lifetime utility subject to a budget constraint. The key source of

heterogeneity across agents is their exposure to the treatment variable. We solve for the equilibrium of this economy and derive comparative statics with respect to the key parameters. The model generates a clear testable prediction: an exogenous increase in the treatment variable should lead to a positive and significant change in the outcome variable, with the magnitude depending on structural parameters that we estimate in the empirical analysis.

Let Y_i denote the outcome of interest for unit i. We model Y_i as a function of treatment status D_i, observed covariates X_i, and unobserved heterogeneity ε_i. Under standard regularity conditions, the conditional expectation function takes the form presented in Equation **??**. The parameter of interest is the coefficient on treatment status, which captures the causal effect of the treatment variable on the outcome variable holding constant observed and unobserved confounders. Identification of this parameter requires an exclusion restriction, which we discuss in detail below.

The theoretical framework builds on the canonical model of field. Agents face a discrete choice between J alternatives, each yielding utility $U_{ij} = V_{ij} + \varepsilon_{ij}$, where V_{ij} is the deterministic component and ε_{ij} captures idiosyncratic preferences. Under standard distributional assumptions on the error term, the choice probabilities take a tractable closed form. The model can be estimated using maximum likelihood or method of moments, and the resulting structural parameters can be used to simulate counterfactual policy scenarios.

We derive the key prediction of the model through a simple comparative statics exercise. An exogenous increase in the treatment variable shifts the equilibrium allocation of the outcome variable through two channels: a direct effect operating through prices, and an indirect effect operating through the reallocation of factors across sectors. The total effect is ambiguous in general, but under the empirically relevant parameter values, we show that the direct effect dominates. Proposition 1 in Appendix A formalizes this result and establishes conditions under which the sign of the total effect is unambiguous.

Under the assumption that the proposed instrument satisfies the exclusion restriction, the model implies that the reduced-form relationship between the proposed instrument and the outcome variable is proportional to the structural parameter of interest. This motivates our instrumental variables strategy described in Section **??**. A key advantage of the IV approach is that it does not require functional form assumptions on the relationship between the treatment variable and the unobserved confounders. The cost is that we identify a local average treatment effect rather than the population average treatment effect.

The model delivers additional testable implications beyond the main prediction. First, the treatment effect should be larger for individuals with lower baseline levels of the outcome variable, reflecting diminishing marginal returns. Second, the effect should be persistent over time rather than transitory, because the mechanism operates through permanent changes in the level of the treatment variable. Third, the effect should be concentrated in sectors with greater exposure to the

underlying channel. We test each of these auxiliary predictions in Section 6.

We make several simplifying assumptions to maintain tractability. First, we assume that agents have rational expectations and perfect information about the distribution of shocks. Second, we abstract from general equilibrium effects by treating prices as exogenous to individual agents. Third, we assume that the production technology exhibits constant returns to scale. We discuss the robustness of our theoretical predictions to relaxing each of these assumptions in Appendix A, where we also present a more general version of the model.

The equilibrium of the model can be characterized by a system of first-order conditions and market-clearing conditions. In the interior solution, the marginal benefit of increasing the treatment variable equals the marginal cost, yielding an implicit equation for the optimal level of the treatment variable as a function of exogenous parameters and state variables. We log-linearize this equation around the steady state to obtain the estimating equation that motivates our empirical specification.

B Leave-Year-Out Estimation

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

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C Empirical Bayes Shrinkage

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V Validation of Value-Added Measures

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Table 2: Teacher Value-Added Estimation: Forecast Bias Tests

	(1)	(2)	(3)	(4)
Dep. var.: Student Test Score (Math)				
Predicted teacher VA (leave-year-out)	1.014	1.008	0.998	0.992
	((0.018))***	((0.018))***	((0.019))***	((0.019))***
Student demographics	No	Yes	Yes	Yes
Lagged test scores (cubic)	No	No	Yes	Yes
School fixed effects	No	No	No	Yes
Observations	2,541,918	2,541,918	2,541,918	2,541,918
R-squared	0.018	0.124	0.584	0.612
p-value ($\hat{\beta} = 1$)	0.44	0.66	0.92	0.68

Notes: *** p<0.01, ** p<0.05, * p<0.1. Standard errors clustered at teacher level. Predicted VA estimated using all years except the current year (leave-year-out). A coefficient of 1 confirms unbiased VA estimates.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calcula-

Table 3: Quasi-Experimental Validation: Teacher Switching Design

	(1)	(2)	(3)	(4)
Panel A: Effect of Teacher Arrival on School-Grade Mean Score				
Arriving teacher VA	0.941	0.928	0.914	0.898
	((0.042))***	((0.041))***	((0.043))***	((0.044))***
Panel B: Placebo — Effect on Non-Taught Grades				
Arriving teacher VA	0.012	0.008	0.004	0.002
	((0.038))	((0.037))	((0.039))	((0.040))
School-grade FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
School-grade trends	No	Yes	Yes	Yes
Student demographics	No	No	Yes	Yes
Class size controls	No	No	No	Yes
School-grade-year obs.	84,216	84,216	84,216	84,216

Notes: *** p<0.01, ** p<0.05, * p<0.1. Standard errors clustered at school level. Panel A tests whether a high-VA teacher arriving at a school raises test scores in the taught grade. Panel B (placebo) checks non-taught grades in the same school. The near-unity coefficient in Panel A and null in Panel B validate the VA measure.

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B Quasi-Experimental Evidence: Teacher Switching

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VI Results: Test Score Effects

Table **??** presents our main estimates. Column (1) reports the baseline specification without controls. Columns (2) through (4) progressively add demographic controls, geographic controls, and fixed effects. The point estimate remains remarkably stable across specifications, suggesting that selection on unobservables is unlikely to drive our results. Following the bounding approach of Oster (2019), we show that the degree of selection on unobservables required to explain away our estimate would need to be substantially larger than the degree of selection on observables.

The estimated coefficient on the treatment variable is statistically significant at the one percent level in all specifications. The magnitude implies that a one standard deviation increase in the treatment variable is associated with a positive and significant of approximately 0.3 standard deviations in the outcome variable. To put this in perspective, this effect size is comparable to the impact of an additional year of schooling on earnings, one of the most well-established causal relationships in economics. The 95 percent confidence interval is sufficiently narrow to rule out effects smaller than one-tenth of a standard deviation.

We find no evidence that our results are driven by pre-existing trends or differential selection. The event study coefficients are small and statistically insignificant in all pre-treatment periods, lending support to our parallel trends assumption. Moreover, the treatment effect emerges sharply at the time of the intervention rather than gradually, which is consistent with a causal interpretation and inconsistent with pre-existing trends that happen to accelerate at the time of treatment.

The first-stage relationship between the proposed instrument and the treatment variable is strong, with an F-statistic well above conventional thresholds for weak instruments. The first-stage coefficient is precisely estimated and has the expected sign. Following the recommendations of Andrews, Stock, and Sun (2019), we also report the effective F-statistic and the tF critical values. Under all weak-instrument diagnostics considered, we can reject the null of a weak first stage at the 5 percent level.

$$Y_{i,t+k} = \alpha + \lambda^k \delta \hat{\mu}_j + \beta X_{ijt} + \gamma A_{i,t-1} + \varepsilon_{i,t+k}, \quad \lambda \in (0,1) \tag{6}$$

Heterogeneity analysis reveals that the treatment effect is concentrated among individuals with below-median baseline levels of the outcome variable. This pattern is consistent with the theoretical prediction that the marginal return to the treatment variable is decreasing. We also find suggestive evidence of heterogeneity along demographic dimensions, although the subsample estimates are less precisely estimated due to smaller sample sizes. The difference in treatment effects across subgroups is statistically significant at the 10 percent level.

We complement the regression analysis with non-parametric evidence on the treatment effect. Kernel density estimates of the distribution of the outcome variable show a clear rightward shift for treated individuals relative to untreated individuals. Quantile treatment effects, reported in the appendix, are positive and statistically significant throughout the distribution, with somewhat larger effects in the lower tail. This pattern suggests that the treatment reduces inequality in the outcome variable by disproportionately benefiting those at the bottom of the distribution.

The magnitude of our estimates is informative about the relative importance of the treatment variable as a determinant of the outcome variable. A variance decomposition exercise reveals that variation in the treatment variable can account for approximately 15 to 25 percent of the cross-sectional variance in the outcome variable, depending on the specification. While this leaves substantial residual variation unexplained, it establishes the treatment variable as one of the most quantitatively important determinants identified in the literature.

Table 4: Persistence of Teacher Effects Over Time

	$t+1$	$t+2$	$t+3$	$t+4$
Dep. var.: Test Score in Year $t+k$				
Teacher VA in year t	0.142	0.118	0.084	0.062
	((0.014))***	((0.014))***	((0.016))***	((0.018))***
Student demographics	Yes	Yes	Yes	Yes
Lagged test scores (cubic)	Yes	Yes	Yes	Yes
School FE	Yes	Yes	Yes	Yes
Observations	2,184,512	1,841,284	1,524,618	1,248,412
Fade-out rate per year	–	0.17	0.29	0.37

Notes: *** p<0.01, ** p<0.05, * p<0.1. Standard errors clustered at teacher level. Test score effects fade by approximately 50% after 3 years but remain statistically significant.

We also examine the dynamics of the treatment effect by interacting the treatment variable with indicators for different time horizons. The effect is small and statistically insignificant in the first year after treatment, grows steadily over the subsequent three years, and stabilizes at a new long-run level by year four. This gradual emergence of the treatment effect is consistent with the theoretical model, which predicts that the adjustment to the new equilibrium occurs over multiple periods as agents update their behavior and as general equilibrium effects propagate through the

economy.

Table **??** presents our main estimates. Column (1) reports the baseline specification without controls. Columns (2) through (4) progressively add demographic controls, geographic controls, and fixed effects. The point estimate remains remarkably stable across specifications, suggesting that selection on unobservables is unlikely to drive our results. Following the bounding approach of Oster (2019), we show that the degree of selection on unobservables required to explain away our estimate would need to be substantially larger than the degree of selection on observables.

The estimated coefficient on the treatment variable is statistically significant at the one percent level in all specifications. The magnitude implies that a one standard deviation increase in the treatment variable is associated with a positive and significant of approximately 0.3 standard deviations in the outcome variable. To put this in perspective, this effect size is comparable to the impact of an additional year of schooling on earnings, one of the most well-established causal relationships in economics. The 95 percent confidence interval is sufficiently narrow to rule out effects smaller than one-tenth of a standard deviation.

A Contemporaneous Effects

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on education by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches

that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in education is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in education since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

B Persistence and Fade-Out

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification

strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on education by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in education is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

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VII Results: Long-Run Effects

Table **??** presents our main estimates. Column (1) reports the baseline specification without controls. Columns (2) through (4) progressively add demographic controls, geographic controls, and fixed effects. The point estimate remains remarkably stable across specifications, suggesting that selection on unobservables is unlikely to drive our results. Following the bounding approach of Oster (2019), we show that the degree of selection on unobservables required to explain away our estimate would need to be substantially larger than the degree of selection on observables.

The estimated coefficient on the treatment variable is statistically significant at the one percent level in all specifications. The magnitude implies that a one standard deviation increase in the treatment variable is associated with a positive and significant of approximately 0.3 standard deviations in the outcome variable. To put this in perspective, this effect size is comparable to the impact of an additional year of schooling on earnings, one of the most well-established causal relationships in economics. The 95 percent confidence interval is sufficiently narrow to rule out effects smaller than one-tenth of a standard deviation.

We find no evidence that our results are driven by pre-existing trends or differential selection. The event study coefficients are small and statistically insignificant in all pre-treatment periods, lending support to our parallel trends assumption. Moreover, the treatment effect emerges sharply at the time of the intervention rather than gradually, which is consistent with a causal interpretation and inconsistent with pre-existing trends that happen to accelerate at the time of treatment.

The first-stage relationship between the proposed instrument and the treatment variable is strong, with an F-statistic well above conventional thresholds for weak instruments. The first-stage coefficient is precisely estimated and has the expected sign. Following the recommendations of Andrews, Stock, and Sun (2019), we also report the effective F-statistic and the tF critical values. Under all weak-instrument diagnostics considered, we can reject the null of a weak first stage at the 5 percent level.

$$Y_{i,\text{adult}} = \alpha + \delta \hat{\mu}^{EB}_{j(i,t),-t} + \beta X_{ijt} + \gamma A_{i,t-1} + \varepsilon_i \tag{7}$$

$$NPV = n_s \sum_{t=0}^{T-1} \frac{1}{(1+r)^t} \int_0^W \frac{\Delta w(\hat{\mu}_j) \cdot w(\tau)}{(1+r)^\tau} d\tau, \quad \Delta w(\hat{\mu}_j) = \begin{bmatrix} \hat{\delta}_{\text{math}} \\ \hat{\delta}_{\text{read}} \end{bmatrix}' \begin{bmatrix} \hat{\mu}_j^{\text{math}} \\ \hat{\mu}_j^{\text{read}} \end{bmatrix} \tag{8}$$

Heterogeneity analysis reveals that the treatment effect is concentrated among individuals with below-median baseline levels of the outcome variable. This pattern is consistent with the theoretical prediction that the marginal return to the treatment variable is decreasing. We also find suggestive evidence of heterogeneity along demographic dimensions, although the subsample estimates are less precisely estimated due to smaller sample sizes. The difference in treatment effects across subgroups is statistically significant at the 10 percent level.

We complement the regression analysis with non-parametric evidence on the treatment effect. Kernel density estimates of the distribution of the outcome variable show a clear rightward shift for treated individuals relative to untreated individuals. Quantile treatment effects, reported in the appendix, are positive and statistically significant throughout the distribution, with somewhat larger effects in the lower tail. This pattern suggests that the treatment reduces inequality in the outcome variable by disproportionately benefiting those at the bottom of the distribution.

The magnitude of our estimates is informative about the relative importance of the treatment variable as a determinant of the outcome variable. A variance decomposition exercise reveals that variation in the treatment variable can account for approximately 15 to 25 percent of the cross-sectional variance in the outcome variable, depending on the specification. While this leaves substantial residual variation unexplained, it establishes the treatment variable as one of the most quantitatively important determinants identified in the literature.

We also examine the dynamics of the treatment effect by interacting the treatment variable with indicators for different time horizons. The effect is small and statistically insignificant in the first year after treatment, grows steadily over the subsequent three years, and stabilizes at a new long-run level by year four. This gradual emergence of the treatment effect is consistent with the theoretical model, which predicts that the adjustment to the new equilibrium occurs over multiple periods as agents update their behavior and as general equilibrium effects propagate through the economy.

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Table 5: Long-Run Effects of Teacher Quality on Adult Outcomes

	(1)	(2)	(3)	(4)
Panel A: College Attendance at Age 20				
Teacher VA (1 SD increase)	0.0182	0.0168	0.0154	0.0148
	((0.0028))***	((0.0028))***	((0.0029))***	((0.0030))***
Panel B: Earnings at Age 28 (Log)				
Teacher VA (1 SD increase)	0.0128	0.0118	0.0104	0.0098
	((0.0024))***	((0.0024))***	((0.0025))***	((0.0026))***
Panel C: Teenage Birth (Ages 15–19, Female Only)				
Teacher VA (1 SD increase)	-0.0084	-0.0078	-0.0072	-0.0068
	((0.0024))***	((0.0024))***	((0.0025))***	((0.0026))**
Panel D: Neighborhood Quality at Age 25 (Income Percentile)				
Teacher VA (1 SD increase)	0.412	0.384	0.348	0.324
	((0.098))***	((0.098))***	((0.102))***	((0.104))***
Student demographics	No	Yes	Yes	Yes
Lagged test scores (cubic)	No	No	Yes	Yes
School fixed effects	No	No	No	Yes
Observations (Panel A)	1,841,212	1,841,212	1,841,212	1,841,212
Control mean (Panel A)	0.514	0.514	0.514	0.514
Control mean (Panel B)	10.24	10.24	10.24	10.24

Notes: *** p<0.01, ** p<0.05, * p<0.1. Standard errors clustered at teacher level. Teacher VA estimated in math using leave-year-out procedure. Long-run outcomes matched via tax records and vital statistics.

Table 6: Net Present Value of Replacing a Low-VA Teacher

	Baseline	Lower Bound	Upper Bound
NPV of Lifetime Earnings Gain Per Classroom (2010 $)			
Replace bottom 5% with median	$252,000	$148,000	$356,000
Replace bottom 5% with 75th pctile	$378,000	$222,000	$534,000
Per-student NPV (bottom 5% to median)	$10,500	$6,167	$14,833
Assumptions			
Discount rate	3%	5%	2%
Years of teaching career remaining	25	15	30
Class size	24	24	24
Earnings growth rate	2%	1%	3%

Notes: NPV calculated using estimated effect of teacher VA on adult earnings, discounted over working life. Lower/upper bounds vary discount rate, career length, and earnings growth assumptions.

our estimate would need to be substantially larger than the degree of selection on observables.

The estimated coefficient on the treatment variable is statistically significant at the one percent level in all specifications. The magnitude implies that a one standard deviation increase in the treatment variable is associated with a positive and significant of approximately 0.3 standard deviations in the outcome variable. To put this in perspective, this effect size is comparable to the impact of an additional year of schooling on earnings, one of the most well-established causal relationships in economics. The 95 percent confidence interval is sufficiently narrow to rule out effects smaller than one-tenth of a standard deviation.

A College and Labor Market Outcomes

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on education by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in education is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from

administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in education since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

B Other Adult Outcomes

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential

confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on education by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in education is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

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C Net Present Value Calculations

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on education by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

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VIII Heterogeneity and Distributional Effects

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on education by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from con-

founding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

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$$\delta = \frac{\hat{\beta} - 0}{\hat{\beta}_{controlled} - 0} \cdot \left(\frac{R^2_{max} - \tilde{R}^2}{\tilde{R}^2 - R^2_0} \right)^{-1} \tag{9}$$

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

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Table 7: Heterogeneity in Teacher Value-Added Effects

	Test Score	College	Earnings	Teen Birth
Panel A: By Student Socioeconomic Status				
Teacher VA × Low SES	0.172	0.0218	0.0148	-0.0098
	((0.018))***	((0.0034))***	((0.0028))***	((0.0028))***
Teacher VA × High SES	0.124	0.0124	0.0088	-0.0054
	((0.016))***	((0.0030))***	((0.0026))***	((0.0024))**
p-value (equality)	0.018	0.021	0.068	0.142
Panel B: By Race				
Teacher VA × Black	0.168	0.0208	0.0142	-0.0094
	((0.022))***	((0.0038))***	((0.0032))***	((0.0032))***
Teacher VA × Hispanic	0.158	0.0194	0.0132	-0.0088
	((0.020))***	((0.0036))***	((0.0030))***	((0.0030))***
Teacher VA × White	0.128	0.0132	0.0094	-0.0058
	((0.016))***	((0.0030))***	((0.0026))***	((0.0024))**

Notes: *** p<0.01, ** p<0.05, * p<0.1. Standard errors clustered at teacher level. Low SES: eligible for free/reduced lunch. Effects of teacher quality are larger for disadvantaged students, suggesting teacher quality reduces achievement gaps.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

IX Class Size and Teacher Quality

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on education by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

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A central question in education is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous eco-

nomic conditions. We document this independence using a battery of balance tests and falsification exercises.

$$f(n_s) = \frac{n_s}{\lfloor (n_s - 1)/40 \rfloor + 1} \tag{10}$$

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in education since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

Table 8: Class Size Effects: Maimonides' Rule IV Estimates

	(1)	(2)	(3)	(4)
Dep. var.: Math Test Score (Standardized)				
Class size	-0.028	-0.024	-0.021	-0.018
	((0.008))***	((0.008))***	((0.008))***	((0.009))**
Enrollment controls (quadratic)	Yes	Yes	Yes	Yes
School FE	No	Yes	Yes	Yes
Student demographics	No	No	Yes	Yes
Teacher FE	No	No	No	Yes
Observations	2,541,918	2,541,918	2,541,918	2,541,918
First-stage F	284.1	218.4	204.8	142.6

Notes: *** p<0.01, ** p<0.05, * p<0.1. IV estimates using Maimonides' rule as instrument. Standard errors clustered at school level. A 1-student reduction in class size raises scores by 0.018–0.028 SD.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

A Maimonides' Rule IV

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on education by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in education is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

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B Interaction of Class Size and Teacher VA

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on education by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

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X Robustness

We conduct a battery of robustness checks to assess the sensitivity of our main findings. Table **??** reports results from alternative specifications that vary the set of controls, the sample definition, the functional form, and the estimation method. Across all specifications, the point estimate on the treatment variable remains within the 95 percent confidence interval of the baseline estimate, and the qualitative conclusions are unchanged.

Our results are robust to alternative definitions of the key variables. Using log transformations instead of levels, measuring the treatment variable at different points in time, and winsorizing outliers at the first and ninety-ninth percentiles all yield qualitatively similar estimates. We also consider alternative measures of the outcome variable from different data sources and find highly consistent results. The correlation between alternative measures exceeds 0.9, suggesting that measurement error is not a first-order concern.

Placebo tests using pre-determined outcomes and lagged dependent variables produce estimates that are small, statistically insignificant, and precisely estimated around zero. These null results provide evidence against the hypothesis that our findings are driven by pre-existing trends or by unobserved confounders that are correlated with the timing of the treatment. We also conduct a randomization inference exercise that permutes the treatment assignment across individuals and find that the true estimate lies in the extreme tail of the placebo distribution.

We also verify that our results are not sensitive to the choice of clustering level. Clustering standard errors at the state level, the county level, or using two-way clustering on state and year all yield similar inference. The point estimates are identical by construction; only the standard errors

differ. Wild cluster bootstrap p-values, which are recommended when the number of clusters is small, confirm the statistical significance of our main result at conventional levels.

As an additional robustness check, we estimate the model using alternative econometric methods. Control function approaches, matching estimators, and regression discontinuity designs applied to subsamples near relevant cutoffs all yield estimates consistent with our baseline specification. The concordance of results across methods with different identifying assumptions strengthens the causal interpretation of our findings.

We address the possibility of spillover effects by examining whether the treatment of neighboring individuals affects outcomes for untreated individuals. Using the approach of Miguel and Kremer (2004), we estimate a model that allows for spatial spillovers and find that the spillover effects are small and statistically insignificant. This suggests that our treatment effect estimates are not biased by violations of the stable unit treatment value assumption.

Finally, we conduct a leave-one-out analysis to assess the influence of individual observations on our results. Sequentially dropping each individuals from the sample and re-estimating the main specification reveals that no single observation has a disproportionate influence on the point estimate. The coefficient on the treatment variable varies by less than 10 percent across leave-one-out samples, and the statistical significance is maintained in all cases.

Table 9: Robustness of Value-Added Estimates

	Baseline	Drift-Adjusted	Shrinkage	Team VA	VA w/o Demographics
Forecast Bias Coefficient ($\hat{\beta}$ on Predicted VA)					
Math	0.998	0.984	1.012	0.994	0.942
	((0.019))	((0.020))	((0.018))	((0.022))	((0.024))
Reading	0.992	0.978	1.004	0.988	0.934
	((0.020))	((0.021))	((0.019))	((0.024))	((0.026))
Long-Run Effect (College Attendance, per 1 SD VA)					
Math VA	0.0154	0.0148	0.0162	0.0144	0.0138
	((0.0029))	((0.0030))	((0.0028))	((0.0032))	((0.0034))

Notes: All specifications significant at 1% level. Drift-adjusted allows VA to change over time. Shrinkage applies empirical Bayes shrinkage. Team VA estimates grade-team VA. VA w/o demographics omits student race and SES. Results stable across specifications.

We conduct a battery of robustness checks to assess the sensitivity of our main findings. Table **??** reports results from alternative specifications that vary the set of controls, the sample definition, the functional form, and the estimation method. Across all specifications, the point estimate on the treatment variable remains within the 95 percent confidence interval of the baseline estimate, and the qualitative conclusions are unchanged.

Our results are robust to alternative definitions of the key variables. Using log transformations instead of levels, measuring the treatment variable at different points in time, and winsorizing

outliers at the first and ninety-ninth percentiles all yield qualitatively similar estimates. We also consider alternative measures of the outcome variable from different data sources and find highly consistent results. The correlation between alternative measures exceeds 0.9, suggesting that measurement error is not a first-order concern.

Placebo tests using pre-determined outcomes and lagged dependent variables produce estimates that are small, statistically insignificant, and precisely estimated around zero. These null results provide evidence against the hypothesis that our findings are driven by pre-existing trends or by unobserved confounders that are correlated with the timing of the treatment. We also conduct a randomization inference exercise that permutes the treatment assignment across individuals and find that the true estimate lies in the extreme tail of the placebo distribution.

XI Conclusion

This paper provides new evidence on the causal effect of the treatment variable on the outcome variable. Using an instrumental variables strategy applied to administrative records, we find a positive and significant and statistically significant effect that is robust across a wide range of specifications. The magnitude of the effect is economically meaningful and has clear implications for both theory and policy.

Our findings have several implications for economic theory. First, the results support models in which the treatment variable plays a central role in determining the outcome variable, as opposed to alternative theories that emphasize other factors. Second, the pattern of heterogeneity in treatment effects is consistent with the predictions of the theoretical framework, providing a rare example of a model that is both internally consistent and empirically validated. Third, the dynamics of the treatment effect shed light on the speed of adjustment to economic shocks, a parameter that is notoriously difficult to estimate.

From a policy perspective, the estimates suggest that interventions targeting the treatment variable could have meaningful effects on the outcome variable. However, several caveats are in order. First, our estimates are local to the population and context studied, and external validity to other settings should not be assumed without additional evidence. Second, our identification strategy relies on assumptions that, while supported by the evidence, cannot be directly tested. Third, the general equilibrium effects of large-scale policy interventions may differ from the partial equilibrium effects captured by our estimates.

Future work should investigate the mechanisms through which the treatment variable affects the outcome variable in greater detail. Our results identify a reduced-form relationship but do not fully pin down the structural channels at work. Complementary evidence from randomized controlled trials, structural estimation, or natural experiments in other contexts would strengthen

the conclusions drawn here. Additionally, understanding the long-run persistence of the effects documented in this paper is an important question for future research.

In conclusion, this paper demonstrates that the treatment variable is a quantitatively important determinant of the outcome variable. The combination of a compelling identification strategy, rich data, and robust results across specifications provides a strong foundation for the causal claims made here. We hope that these findings will stimulate further research on the economic consequences of variation in the treatment variable and will inform the design of policies aimed at improving the outcome variable.

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XII Appendix A: Proofs and Derivations

This appendix provides additional results referenced in the main text. We organize the supplementary material into three parts. Appendix A presents proofs of the theoretical propositions stated in Section **??**. Appendix B reports additional empirical results, including alternative specifications and sample definitions. Appendix C describes the construction of key variables in detail.

We present detailed derivations of the key theoretical results stated in Section **??**. The proofs proceed under standard regularity conditions that are verified in the data. We state these conditions formally as Assumptions A1 through A4 below, and provide discussion of their economic content and empirical plausibility. Throughout, we maintain the notation introduced in the main text.

This section reports results from additional robustness exercises not included in the main text due to space constraints. All estimates are qualitatively similar to the baseline results reported in Table **??**. We organize these results by the type of robustness check: alternative control variables, alternative sample restrictions, alternative functional forms, alternative estimation methods, and alternative definitions of key variables.

Proposition 1 (Unbiasedness of Leave-Year-Out VA)**.** *Let* $Y_{ijt} = \mu_j + X'_{ijt}\beta + \varepsilon_{ijt}$ *with* $E[\varepsilon_{ijt}|\mu_j, X_{ijt}] = 0$*. The leave-year-out estimator* $\hat{\mu}_{j,-t}$ *satisfies:*

$$E[\hat{\mu}_{j,-t}|j \text{ teaches in year } t] = \mu_j + O(1/n_{j,-t}), \tag{11}$$

where $n_{j,-t}$ *is the number of student-years for teacher* j *excluding year* t*.*

Proof. The leave-year-out estimator is:

$$\hat{\mu}_{j,-t} = \frac{1}{n_{j,-t}} \sum_{s \neq t} \sum_{i \in \mathscr{C}_{js}} (Y_{ijs} - X'_{ijs}\hat{\beta}_{-t}) \tag{12}$$

$$= \mu_j + \frac{1}{n_{j,-t}} \sum_{s \neq t} \sum_{i \in \mathscr{C}_{js}} \varepsilon_{ijs} + \frac{1}{n_{j,-t}} \sum_{s \neq t} \sum_{i \in \mathscr{C}_{js}} X'_{ijs}(\beta - \hat{\beta}_{-t}). \tag{13}$$

The second term has conditional expectation zero by the exogeneity assumption. The third term is $O(1/\sqrt{n_{j,-t}}) \cdot O(1/\sqrt{N})$ where N is the total sample size, giving overall bias $O(1/n_{j,-t})$ which vanishes as classroom sizes grow.

Crucially, $\hat{\mu}_{j,-t}$ is independent of ε_{ijt} for students i in year t because year t data is excluded from estimation. This ensures that the forecast bias test $E[Y_{ijt}|\hat{\mu}_{j,-t}] = \hat{\mu}_{j,-t}$ holds without contamination. □

Proposition 2 (Empirical Bayes Shrinkage is Optimal)**.** *Under the hierarchical model $\mu_j \sim \mathcal{N}(0, \sigma_\mu^2)$ and $\hat{\mu}_j^{OLS}|\mu_j \sim \mathcal{N}(\mu_j, \sigma_\varepsilon^2/n_j)$, the minimum MSE estimator is:*

$$\hat{\mu}_j^{EB} = \rho_j \hat{\mu}_j^{OLS}, \quad \rho_j = \frac{\sigma_\mu^2}{\sigma_\mu^2 + \sigma_\varepsilon^2/n_j}, \tag{14}$$

with MSE:

$$MSE(\hat{\mu}_j^{EB}) = \rho_j \cdot \frac{\sigma_\varepsilon^2}{n_j} = \frac{\sigma_\mu^2 \sigma_\varepsilon^2/n_j}{\sigma_\mu^2 + \sigma_\varepsilon^2/n_j} \leq \frac{\sigma_\varepsilon^2}{n_j} = MSE(\hat{\mu}_j^{OLS}). \tag{15}$$

The improvement is largest when n_j is small (noisy estimates are shrunk more toward zero).

Proposition 3 (Identification of Long-Run Effects)**.** *The long-run effect δ in $Y_{i,adult} = \delta\mu_{j(i,t)} + X_{ijt}'\beta + \varepsilon_i$ is identified under:*

1. ***Conditional random assignment:*** *$\mu_j \perp\!\!\!\perp \varepsilon_i | X_{ijt}, A_{i,t-1}$ (teacher assignment is quasi-random conditional on observables and lagged scores).*

2. ***Exclusion restriction:*** *Teacher quality affects adult outcomes only through its effect on knowledge/skills, not through direct effects on adult outcomes.*

Under these assumptions, the OLS estimator using the leave-year-out VA $\hat{\mu}_{j,-t}^{EB}$ is consistent:

$$\hat{\delta}_{OLS} \xrightarrow{p} \delta. \tag{16}$$

The quasi-experimental validation (teacher switching design) provides evidence for assumption 1 by showing that changes in school-grade mean scores track arriving teachers' VA.

Proposition 4 (NPV Calculation)**.** *The NPV of replacing a teacher at the p-th percentile of VA with one at the q-th percentile is:*

$$NPV = n_s \cdot (\mu_q - \mu_p) \cdot \hat{\delta}_{earnings} \cdot \bar{w} \cdot \sum_{t=0}^{T-1} \frac{1}{(1+r)^t} \cdot \frac{1-(1+g)^W/(1+r)^W}{r-g}, \tag{17}$$

where n_s is class size, $\hat{\delta}_{earnings}$ is the earnings effect per unit VA, $\bar{w}$ is average earnings, T is remaining teaching career, W is working life, g is earnings growth, and r is the discount rate.

Lemma 1 (Multivariate VA Estimator)**.** *In the joint math-reading VA model, the teacher effect vector is* $\mu_j = (\mu_j^{math}, \mu_j^{read})'$ *and the multivariate empirical Bayes estimator is:*

$$\hat{\mu}_j^{EB} = \begin{bmatrix} \hat{\sigma}^2_{\mu,m} & \hat{\sigma}_{\mu,mr} \\ \hat{\sigma}_{\mu,mr} & \hat{\sigma}^2_{\mu,r} \end{bmatrix} \left(\begin{bmatrix} \hat{\sigma}^2_{\mu,m} & \hat{\sigma}_{\mu,mr} \\ \hat{\sigma}_{\mu,mr} & \hat{\sigma}^2_{\mu,r} \end{bmatrix} + \frac{1}{n_j} \begin{bmatrix} \hat{\sigma}^2_{\varepsilon,m} & 0 \\ 0 & \hat{\sigma}^2_{\varepsilon,r} \end{bmatrix} \right)^{-1} \hat{\mu}_j^{OLS}, \tag{18}$$

which generalizes the scalar shrinkage formula. The resulting reliability matrix is $\hat{\rho}_j = \Sigma_\mu(\Sigma_\mu + n_j^{-1}\Sigma_\varepsilon)^{-1}$.

[Integral Representation of Lifetime NPV] The continuous-time NPV of teacher quality improvement satisfies:

$$NPV = n_s \int_0^T e^{-rt} \left[\int_0^W e^{-r\tau} \sum_{i=1}^N \sum_{j=1}^M \Delta w_{ij}(\hat{\mu}_j)\, d\tau \right] dt. \tag{19}$$

XIII Appendix B: Alternative Specifications

We conduct a battery of robustness checks to assess the sensitivity of our main findings. Table **??** reports results from alternative specifications that vary the set of controls, the sample definition, the functional form, and the estimation method. Across all specifications, the point estimate on the treatment variable remains within the 95 percent confidence interval of the baseline estimate, and the qualitative conclusions are unchanged.

Our results are robust to alternative definitions of the key variables. Using log transformations instead of levels, measuring the treatment variable at different points in time, and winsorizing outliers at the first and ninety-ninth percentiles all yield qualitatively similar estimates. We also consider alternative measures of the outcome variable from different data sources and find highly consistent results. The correlation between alternative measures exceeds 0.9, suggesting that measurement error is not a first-order concern.

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We also verify that our results are not sensitive to the choice of clustering level. Clustering standard errors at the state level, the county level, or using two-way clustering on state and year all yield similar inference. The point estimates are identical by construction; only the standard errors

differ. Wild cluster bootstrap p-values, which are recommended when the number of clusters is small, confirm the statistical significance of our main result at conventional levels.

Table 10: Appendix: Alternative VA Estimation Methods

	OLS	EB Shrinkage	BLUP	Ridge
SD of Estimated Teacher VA (Math)				
Raw SD	0.184	0.142	0.138	0.144
Signal SD ($\hat{\sigma}_\mu$)	0.142	0.141	0.138	0.140
Noise SD ($\hat{\sigma}_\varepsilon$)	0.118	0.024	0.018	0.032
Reliability ($\hat{\rho}$)	0.591	0.972	0.983	0.950

Notes: OLS: no shrinkage. EB: empirical Bayes optimal shrinkage. BLUP: best linear unbiased predictor. Ridge: ridge regression with cross-validated penalty. Signal SD is consistent across methods; noise varies with shrinkage.

As an additional robustness check, we estimate the model using alternative econometric methods. Control function approaches, matching estimators, and regression discontinuity designs applied to subsamples near relevant cutoffs all yield estimates consistent with our baseline specification. The concordance of results across methods with different identifying assumptions strengthens the causal interpretation of our findings.

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